

TMX-063 — PRODUCTION

root-free zig build: captured evidence

Revision: 1 **Last modified:** 2026-06-29T16:30:00Z **Authority:** vasic-digital tmux — QA evidence trail (§11.4.83) **Scope:** the PROVEN PoC (verdict A) wired into the production scripts — scripts/obtain_local_deps.sh + scripts/build_native.sh + scripts/setup.sh build tmux 3.6a with NO root, NO sudo/su, NO interaction, even when the host has NO working C toolchain (obtained zig 0.16.0 cc toolchain).

Verdict

WORKS via the unmodified project scripts. On a real host (nezha, Linux x86_64) the REAL scripts/obtain_local_deps.sh + scripts/build_native.sh, run inside a neutered env (no cc/gcc/clang/ld/as/ar/autoconf/automake/bison/ pkg-config reachable), obtained the zig toolchain, source-built libevent + ncurses + jemalloc with it, and built tmux 3.6a from the 3.6a RELEASE tarball — tmux -V == tmux 3.6a, a live session ran, N=3 builds were identical, and the §1.1 mutation was caught. NORMAL-host build path unchanged (no regression). RED→GREEN polarity proven.

Test results (scripts/tests/71_root_free_zig_build.sh)

- RED_MODE=1 (reproduce the defect on the broken artifact): **PASS=5 FAIL=0 SKIP=0** — see test71_RED_full.log. C3 proves the neutered build WITHOUT the zig obtain FAILS at the can't-link step (defect present).
- RED_MODE=0 (standing GREEN guard): **PASS=12 FAIL=0 SKIP=0** — see test71_GREEN_full.log.
- Sibling regression scripts/tests/70_native_fallback_cc_link.sh: **PASS=16 FAIL=0 SKIP=0** (my setup.sh edits introduce no regression).

Captured proofs

Proof	File
C1 NORMAL-host no-regression — cc resolved HOST (CC_KIND=host)	C1_normal_host_resolved.env
C2 neuter validity — host toolchain unreachable	C2_neuter_proof.log
C4/C4b GREEN — CC_KIND=zig (local-toolchain) drove the build; all deps local-build	GREEN_resolved.env
C5 tmux -V == tmux 3.6a (user-visible) + ELF	GREEN_tmux_runtime_proof.log
C6 live session — marker captured from a real pane	GREEN_tmux_live_session.log
C7 residual #2 — DT_NEEDED = LOCAL libncursesw.so.6 (not host libtinfo+libncurses); RUNPATH=local	GREEN_readelf_needed.log
C8 §11.4.50 determinism — N=3 identical tmux 3.6a	GREEN_determinism_n3.log
C3 RED — neutered build with no toolchain fails to link	RED_no_toolchain_build_fail.log

Residuals fixed

1. **obtain_via_source bypassed the obtained zig** — fixed: obtain_via_source now consumes RESOLVED_CC_KIND=zig (CC = the flag-filter wrapper, wrapper dir prepended to PATH so ar/ranlib/objcopy/ld resolve to zig), and resolve_cc resolves the host compiler by LINK-CAPABILITY (a host cc that cannot link falls through to OBTAIN). MAKE is pinned to the absolute path so config.status
 - recursive \$(MAKE) work on a bare/neutered PATH.
2. **ncurses host-preference** — fixed: _ncurses_compat_symlinks creates libtinfo.so/libtinfo.so.6/libncurses.so/libncurses.so.6 → the local widec libncursesw.so.6 (which bundles setupterm), so tmux's AC_SEARCH_LIBS(setupterm,[tinfo terminfo ncurses ncursesw]) links the LOCAL copy first (-L<local> precedes /lib64). Proven by GREEN_readelf_needed.log showing DT_NEEDED libncursesw.so.6, not the host libtinfo.so.6+libncurses.so.6.

Proven payload

- zig 0.16.0 x86_64-linux, sha256
70e49664a74374b48b51e6f3fd9bf437f6395d42509050588bd49abe52ba3d0
0 (byte-verified pin == index.json shasum, cross-checked live).
Other 3 tuples' shasums are fetched + parsed from <https://ziglang.org/download/index.json> at obtain time (never
hardcoded unverified).
- tmux 3.6a release tarball sha256
b6d8d9c76585db8ef5fa00d4931902fa4b8cbe8166f528f44fc403961a3f375
9.

Host safety (§12 / §11.4.133)

NO sudo/su, NO root, NO host-toolchain mutation, NO interaction.
The test runs the build into a scratch LOCAL_DEPS_ROOT + scratch
TMX_BUILD_DIR (new env overrides in build_native.sh) so it NEVER
clobbers the operator's tmux/build or .local-deps. CM-NO-SUDO-
NO-INTERACTION + CM-LOCAL-DEPS-MECHANISM gates re-
verified clean against the current tree.

Sources verified 2026-06-29

- zig downloads + index.json shasum schema: <https://ziglang.org/download/index.json>
- ncurses 6.5 tarball (sha-verified): <https://invisible-island.net/archives/ncurses/ncurses-6.5.tar.gz>
- tmux 3.6a release tarball: <https://github.com/tmux/tmux/releases/download/3.6a/tmux-3.6a.tar.gz>
- Full research artefact: docs/research/
root_free_c_toolchain_20260629/README.md